Aashish Dhawan

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SUMMARY

Machine Learning Engineer and PhD researcher with **5+ years** building and deploying ML systems across NLP, computer vision, and multimodal AI. Experienced fine-tuning large language models (LLaMA, GPT-4, mBART-50) with LoRA/PEFT, designing **RAG** pipelines, and shipping production-grade AI at **ExxonMobil** and for 5,000+ end users. Published at **ACL 2026** (shared task winner) and EACL 2026. Seeking Applied Scientist, ML Engineer, or Data Scientist roles.

SKILLS

Languages: Python, SQL, Java, C++, R
ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, XGBoost
LLMs & NLP: GPT-4, LLaMA, Gemini, Qwen2.5-VL, mBART-50, Fine-tuning (LoRA/PEFT), Retrieval-Augmented Generation (RAG), LangChain, Multilingual NLP, Low-Resource MT
Computer Vision: CLIP, SAM, DINO, Object Detection, Image Segmentation, Diffusion Models
Tools & MLOps: FastAPI, AWS, Docker, Git, FAISS, Streamlit, Pandas, NumPy

EXPERIENCE

Computational Data Scientist Intern — **May 2026 – Present**
ExxonMobil — Houston, TX

- Developed Python XAI pipelines that generate counterfactual explanations and feature-level attributions for industrial optimization models, enabling operations teams to interrogate and validate model-driven decisions in production workflows.
- Built what-if simulation tooling allowing stakeholders to perturb model inputs, evaluate sensitivity of optimization outputs, and reason about model behavior without ML expertise—reducing reliance on black-box predictions for high-stakes industrial decisions.

Machine Learning Researcher — **May 2024 – Present**
Data Science & Research Lab, University of Florida — Gainesville, FL

- Engineered a few-shot object detection system (CLIP + SAM + DINO + Adaptive Feature Transfer + OHAC clustering) achieving **15% accuracy improvement** over CLIP baselines in 1–10-shot regimes, with interpretable concept-level visual explanations per prediction.
- Built a production **RAG pipeline** for low-resource Indigenous language translation using Qwen2.5-VL image captioning, BM25 retrieval, and Gemini 2.5 Flash many-shot prompting; system **ranked 1st at AmericasNLP 2026 Shared Task (ACL)**.
- Fine-tuned multilingual **mBART-50** with distributed data-parallel (DDP) multi-GPU training, reducing training time by **50%**; evaluated using chrF++ with full ablation studies published at EACL 2026.

Adjunct Instructor — **Aug 2023 – Present**
University of Florida — Gainesville, FL

- Designed and delivered Machine Learning and Programming Languages courses for **300+ students** per semester; developed assignments, rubrics, and project-based evaluations.
- Mentored capstone projects: LLM-powered chatbots, recommendation systems, and real-time computer vision pipelines.

AI Engineer, Consultant — **Aug 2023 – Apr 2024**
Atmosphere Apps — Gainesville, FL

- Engineered GPT-4 and Whisper-based NLP pipelines for prescription recommendation and multilingual transcription, improving matching accuracy by **20%** for **5,000+** active users.
- Delivered audio/video transcription across 10+ languages at **95% accuracy**, producing structured summaries, topic extractions, and auto-generated quizzes at scale.

Research Assistant — **Jan 2021 – May 2023**
Graphics Imaging & Light Measurement Lab, University of Florida — Gainesville, FL

- Developed deep learning pipelines for multispectral image synthesis, quantum simulation acceleration, and material-scattering prediction, reducing evaluation time by up to **82%**.

Visiting Researcher — **Dec 2019 – Mar 2020**
UBTECH AI Lab, University of Sydney — Sydney, Australia

- Implemented multi-source domain adaptation (CNN backbone, shared encoder + domain-specific classifiers); achieved 71–98% accuracy across 345 categories on a 0.6M-sample dataset.

Research Intern — **May 2019 – Aug 2019**
Space Applications Center, ISRO — Ahmedabad, India

- Built a CNN–CRF semantic segmentation pipeline for satellite imagery (LISS-3 + POTSDAM), improving accuracy by 5% and cutting inference from 36 hours to 0.2 seconds (**648,000× speedup**).

EDUCATION

Ph.D. in Computer Science — **Aug 2025 – May 2028 (Expected)**
University of Florida — Gainesville, FL

M.S. in Computer Science — **Jan 2021 – May 2023**
University of Florida — Gainesville, FL

- **Coursework:** Natural Language Processing, Deep Learning for Computer Graphics, Trustworthy ML

Bachelor of Technology in Computer Science — **Aug 2015 – May 2019**
Kurukshetra University — Haryana, India

PUBLICATIONS

- Dhawan et al. "Retrieval-Augmented Long-Context Translation for Cultural Image Captioning." *AmericasNLP @ ACL 2026.* arXiv:2605.20626. [**Overall Shared Task Winner**]
- Dhawan et al. "Improving Indigenous Language MT with Synthetic Data and Language-Specific Preprocessing." *LoResMT @ EACL 2026.* arXiv:2601.03135.
- Driggers-Ellis, . . . , Dhawan et al. "MultiScript30k: Multilingual Embeddings for Cross-Script Parallel Data." arXiv:2512.11074. *Under review, ACL Rolling Review.*
- Dhawan et al. "Post Processing of Image Segmentation Using CRF." *IEEE INDIACom*, 2019.

SELECTED PROJECTS

Explainable Few-Shot Object Detection (Indigenous Manuscripts) | *PyTorch, CLIP, SAM, DINO, FastAPI* — **2025**

- Concept-based few-shot detection for Mixtec manuscripts using CLIP, SAM, DINO, and Adaptive Feature Transfer; **15% accuracy gain** over CLIP with hierarchical visual explanations per prediction served via a FastAPI inference backend.

RAGFlow: Documentation Assistant | *Python, FastAPI, OpenAI API, FAISS, LangChain* — **2025**

- Built a **retrieval-augmented generation (RAG)** assistant using FAISS vector search, semantic chunking, and GPT-based response generation; served via FastAPI with sub-second retrieval latency.

Autonomous Job Application Agent | *Python, FastAPI, Gemini, PostgreSQL, Streamlit* — **2025**

- End-to-end AI agent that scrapes job listings, scores role fit with **Gemini**, generates tailored cover letters via LLM prompting, and automates application submissions through a PostgreSQL-backed pipeline.